Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years shipping **AI-powered features**, **React/TypeScript interfaces**, and **data-driven applications**. Experienced building with LLMs (GPT-4, Claude), iterating rapidly in startup environments, and translating user feedback into shipped products.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Shipped **AI-native features using GPT-4o, Claude, and Qwen-VL** with switchable model architecture, building natural language workflows that processed 10K+ daily requests and reduced feedback turnaround by **60%**.
- Built **React interfaces and Python APIs** in a fast-moving startup, iterating rapidly on user feedback from 150+ instructors while maintaining stable performance under production workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **data analysis pipelines processing 100K+ records** with PyTorch, implementing feature extraction, visualization workflows, and experiment tracking for reproducible analytics and model evaluation.
- Designed **transformer-based models** with multi-head attention, improving planning accuracy by 18% through rapid experimentation, systematic benchmarking, and iterative model refinement.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **analytics dashboard backend** serving 500+ users, reducing query latency from **1.2s to 350ms** through database optimization, enabling faster data exploration and real-time insights.
- Developed **data pipelines** processing 5K+ monthly jobs with automated validation, supporting analytics workflows with 99%+ reliability across production deployments.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **real-time collaboration platform** with MERN stack and WebSockets, enabling 2M+ record sync across distributed users with live updates and conflict resolution.
- Shipped features rapidly with **Docker-based CI/CD**, cutting release cycles by **50%** through automated testing, staged deployments, and quick iteration on user feedback.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **real-time analytics with Databricks and Flask**, enabling data analysis on 50K+ records with sub-200ms API responses and Firebase-powered live synchronization.
- Developed **cross-platform React interfaces** for web and mobile, translating product requirements directly into shipped features while gathering user feedback to guide iteration.

Skills

Frontend: **React**, **TypeScript**, Next.js, JavaScript, responsive UI, data visualization
AI/LLM: GPT-4/Claude APIs, LLM orchestration, AI-assisted development, coding agents, prompt engineering
Backend: Python, Flask, FastAPI, Node.js, REST APIs, PostgreSQL, MongoDB
Data: Data pipelines, analytics workflows, Databricks, PyTorch, feature engineering
Tools: Git, Docker, AWS, CI/CD, rapid iteration, Agile/startup workflows

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built an **AI-powered data intelligence platform** with natural language search and Q&A across 50K+ documents, achieving 95% query accuracy through vector embeddings and LLM-powered analysis.

PiXMod Live | *Next.js, TypeScript, MongoDB, Transformers*

- Built an **AI-native SaaS platform** with CLIP and Stable Diffusion, implementing semantic search across 100K+ embeddings with real-time processing and intuitive user workflows.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets*

- Built a **real-time analytics dashboard** supporting 50K+ users and 1M+ transactions with live data sync, advanced filtering, and responsive visualizations for financial insights.